

The converse of Pythagoras’ theorem, and applications

Running the theorem backwards to test for a right angle — and using it on real shapes.

LESSON 22–23

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 18–19

STAGE 9 · 5.5

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the converse of Pythagoras’ theorem and use it to test a triangle.
- Classify a triangle as acute, right or obtuse from its three sides.
- Apply Pythagoras to rectangles, rhombuses, trapeziums and isosceles triangles.
- Solve practical distance problems.

TEXTBOOK

National	Темы 18–19, pp. 44–48
Cambridge	Learner’s Book pp. 124–126
Workbook	Workbook 5.5

01

Explanation

The converse

CONVERSE THEOREM

If the sides of a triangle satisfy $a^2 + b^2 = c^2$, where c is the longest side, then the triangle is **right-angled**, with the right angle opposite c .

The original theorem says “right angle $\implies$ the relation”. The converse says “the relation $\implies$ a right angle”. Both are true here, but that is a piece of luck, not a rule — most theorems have a false converse, which is exactly why the two directions get separate names.

How to use it. Square all three sides. If the two smaller squares add to the largest, the triangle is right-angled.

Classifying any triangle

Comparing $a^2 + b^2$ with c^2 tells you more than just “right or not”:

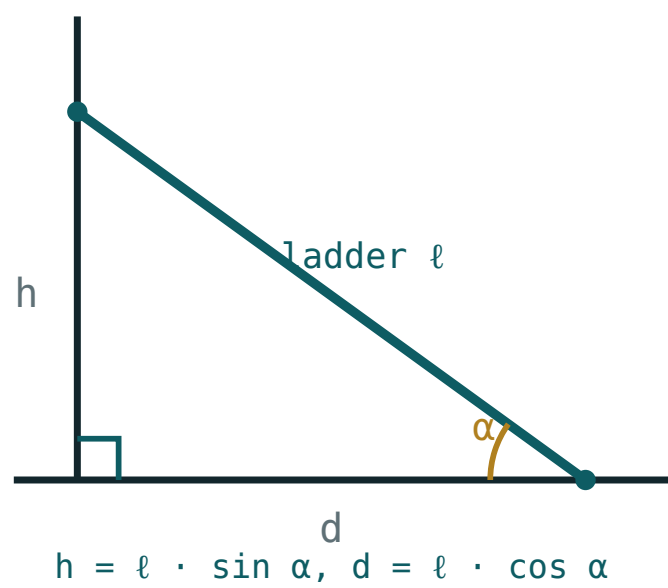
COMPARISON	THE TRIANGLE IS	EXAMPLE
$a^2 + b^2 = c^2$	right-angled	3, 4, 5
$a^2 + b^2 > c^2$	acute-angled	4, 5, 6
$a^2 + b^2 < c^2$	obtuse-angled	2, 3, 4

ALWAYS TEST WITH THE LONGEST SIDE AS C

For 5, 12, 13 the test is $25 + 144 = 169 \checkmark$. Testing $25 + 169$ against 144 proves nothing.

The four standard applications

- **Diagonal of a rectangle.** $d = \sqrt{a^2 + b^2}$.
- **Height of an isosceles triangle.** Drop the height to the base; it bisects the base, so $h = \sqrt{\text{side}^2 - (\text{base}/2)^2}$.
- **Side of a rhombus.** The half-diagonals are perpendicular legs: $a = \sqrt{(d_1/2)^2 + (d_2/2)^2}$.
- **Height of an isosceles trapezium.** Each end triangle has horizontal leg $\frac{a - b}{2}$.



The ladder problem: the wall, the ground and the ladder always form a right triangle.

The practical version is always the same picture — a vertical, a horizontal and a slant.

02 Worked examples

EXAMPLE 1 Is the triangle with sides 8, 15, 17 right-angled?

- 1 *longest side $c = 17$*
Identify it first.

- 2 $8^2 + 15^2 = 64 + 225 = 289$

- 3 $17^2 = 289$

- 4 They are equal, so by the converse the triangle is right-angled.
The right angle is opposite the side 17.

ANSWER

Yes — right-angled.

EXAMPLE 2 Classify the triangle with sides 4, 5, 6.

- 1 $c = 6$
The longest side.

- 2 $4^2 + 5^2 = 16 + 25 = 41$

- 3 $6^2 = 36$

- 4 $41 > 36$
The sum of the smaller squares is larger.

ANSWER

Acute-angled.

EXAMPLE 3 A rhombus has diagonals 10 cm and 24 cm. Find its side and perimeter.

① *half-diagonals 5 and 12*

The diagonals bisect each other.

② They are perpendicular, so they are the legs of a right triangle.

The side of the rhombus is the hypotenuse.

③ $a = \sqrt{25 + 144} = 13 \text{ cm}$

④ $P = 4 \cdot 13 = 52 \text{ cm}$

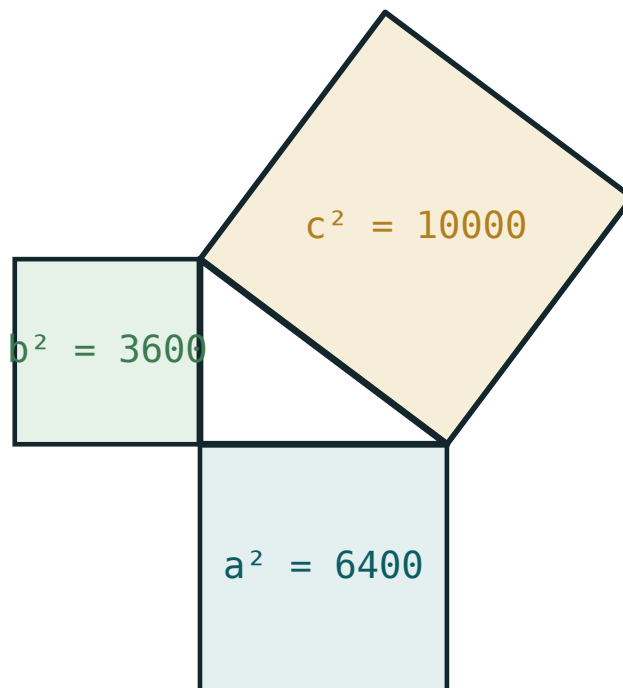
ANSWER

$a = 13 \text{ cm}, P = 52 \text{ cm}$

03 Interactive model

Set the legs to 8 and 15 and read c — then ask the class to name the triple.

Test a triangle for a right angle



a 80

b 60

c 100

a^2 6400

b^2 3600

$a^2 + b^2$ 10000

c^2 10000

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Converse theorem	Teskari teorema	Обратная теорема
Test for a right angle	To'g'ri burchak alomati	Признак прямого угла
Acute triangle	O'tkir burchakli uchburchak	Остроугольный треугольник
Obtuse triangle	O'tmas burchakli uchburchak	Тупоугольный треугольник
Longest side	Eng uzun tomon	Наибольшая сторона
Diagonal	Diagonal	Диагональ
Height	Balandlik	Высота
Distance	Masofa	Расстояние
Apply	Qo'llash	Применить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Sides 9, 40, 41. The triangle is:

right-angled

acute

obtuse

impossible

Sides 2, 3, 4. The triangle is:

right-angled

acute

obtuse

impossible

To test sides 7, 24, 25 you compare:

$7^2 + 25^2$ with 24^2

$7^2 + 24^2$ with 25^2

$24^2 + 25^2$ with 7^2

any of these

A rhombus with diagonals 6 and 8 has side:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Is 3, 4, 5 right-angled?*

ANSWER Yes.

2. *Is 5, 12, 13 right-angled?*

ANSWER Yes.

3. *Is 4, 5, 6 right-angled?*

ANSWER No — it is acute.

4. *Is 2, 3, 4 right-angled?*

ANSWER No — it is obtuse.

5. *A rectangle is 6 by 8. Find its diagonal.*

ANSWER 10

6. *A rhombus has diagonals 6 and 8. Find its side.*

ANSWER 5

7. *Which side must be squared alone in the test?*

ANSWER The longest one.

Medium · the standard question

7 PROBLEMS

1. *Is 8, 15, 17 right-angled?*

ANSWER Yes.

2. *Classify the triangle with sides 4, 5, 6.*

ANSWER Acute.

3. *Classify the triangle with sides 5, 6, 10.*

ANSWER $25 + 36 = 61 < 100$ — obtuse.

4. *A rhombus has diagonals 10 and 24. Find its side and perimeter.*

ANSWER 13, 52

5. *An isosceles triangle has equal sides 10 and base 16. Find its height.*

ANSWER 6

6. *A rectangle has diagonal 25 and one side 7. Find the other side.*

ANSWER 24

7. *A ladder 17 m long has its foot 8 m from a wall. How high does it reach?*

ANSWER 15 m

Hard · stretch and reason

7 PROBLEMS

1. An isosceles trapezium has bases 8 and 20 and legs 10. Find its height and area.

ANSWER $\frac{20 - 8}{2} = 6, h = 8, S = 112$

2. An equilateral triangle has side 10. Find its height and area.

ANSWER $h = 5\sqrt{3}, S = 25\sqrt{3}$

3. A right triangle has perimeter 30 and hypotenuse 13. Find its legs.

ANSWER $a + b = 17$ and $a^2 + b^2 = 169$ give $ab = 60$: legs 5 and 12.

4. Two points are 3 km east and 4 km north of a village. How far apart are they?

ANSWER 5 km

5. Show that 20, 21, 29 is a Pythagorean triple.

ANSWER $400 + 441 = 841 = 29^2$

6. A rhombus has side 25 and one diagonal 14. Find the other diagonal.

ANSWER half-diagonal $\sqrt{625 - 49} = 24$, so the diagonal is 48.

7. For which x is the triangle with sides 6, 8, x right-angled?

ANSWER $x = 10$ (hypotenuse) or $x = 2\sqrt{7}$ (leg).

07 Homework

Homework — 6 problems

Geometry 8, Темы 18–19, pp. 44–48.

1. Is 9, 12, 15 right-angled? Show the test.
2. Classify the triangle with sides 7, 8, 12.
3. A rhombus has diagonals 16 and 30. Find its side and perimeter.
4. An isosceles triangle has equal sides 17 and base 16. Find its height.

5. *A rectangle has diagonal 41 and one side 9. Find the other side and the area.*
6. *A ladder 25 m long reaches 24 m up a wall. How far is its foot from the wall?*